```
  [1] "abc"    "abc.data"   "abtest"   "AMR"
"base64enc"
  [6] "bit"    "bit64"    "blob"      "brio"         "bslib"
 [11] "cachem"  "callr"   "cli"      "commonmark"   "crayon"
 [16] "cubature"  "DBI"   "desc"    "diffobj"      "digest"
 [21] "doParallel"   "ellipsis"     "evaluate"     "fansi"
"fastmap"
 ...
```

Remove

You can remove a package using the *remove.package* function. For example:

```
> remove.packages(c("abc","abc.data", "abtest"),"/home/
guest/R/x86_64-pc-linux-gnu-library/4.1")
```

The second argument specifies the path where the packages are installed by default. If you install the packages in a different directory, you can set it to the *Library* variable, and use it as the second argument to *remove.packages()*. We can again verify that the packages have been removed:

```
> (.packages(all.available=TRUE))
  [1] "AMR"    "base64enc"   "bit"      "bit64"
"blob"
  [6] "brio"   "bslib"     "cachem"       "callr"
"cli"
 [11] "commonmark"   "crayon"  "cubature"  "DBI"
"desc"
 [16] "diffobj"  "digest"  "doParallel"  "ellipsis"
"evaluate"
 [21] "fansi"   "fastmap"   "fontawesome"  "foreach"
"fs"
 ...
```

Update

The *old.packages* function returns a list of packages that have a newer version released in a remote R repository. For example:

```
> old.packages()
        Package    LibPath
Installed
cli      "cli"      "/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1" "3.1.0"
DBI      "DBI"      "/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1" "1.1.1"
fansi    "fansi"    "/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1" "1.0.0"
glue     "glue"     "/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1" "1.4.2"
...
        Built   ReposVer  Repository
cli      "4.1.2" "3.1.1"   "https://cloud.r-project.org/src/
contrib"
DBI      "4.1.0" "1.1.2"   "https://cloud.r-project.org/src/
contrib"
fansi    "4.1.2" "1.0.2"   "https://cloud.r-project.org/src/
contrib"
glue     "4.1.0" "1.6.1"   "https://cloud.r-project.org/src/
contrib"
...
```

The *cli.3.1.0* package has been installed on the system, but there is a newer '3.1.1' package available. We can update the package using the *update.packages* function, as follows:

```
> update.packages()
cli :
 Version 3.1.0 installed in /home/shakthi/R/x86_64-pc-linux-
gnu-library/4.1
 Version 3.1.1 available at https://cloud.r-project.org
Update? (Yes/no/cancel) Yes
DBI :
 Version 1.1.1 installed in /home/shakthi/R/x86_64-pc-linux-
gnu-library/4.1
 Version 1.1.2 available at https://cloud.r-project.org
Update? (Yes/no/cancel) no
...
 ** building package indices
 ** testing if installed package can be loaded from temporary
location
 ** checking absolute paths in shared objects and dynamic
libraries
 ** testing if installed package can be loaded from final
location
 ** testing if installed package keeps a record of temporary
installation path
 * DONE (cli)
```

The command prompts you for a confirmation before proceeding to update the package. You can choose not to update a specific package, or cancel the entire operation. In the above example, we only update the *cli* package to 3.1.1. The same can be verified using the *packageVersion* function, as follows:

```
> packageVersion("cli")
[1] '3.1.1'
```

Download

The *new.packages* function shows the uninstalled packages from the local system that are available in remote R